



MANIPAL UNIVERSITY JAIPUR

School of Information Security & Data Science

DEPARTMENT OF DATA SCIENCE & ENGINEERING

Course Hand-out

Block Chain Technologies | DS4142 | Credits | 3 0 0 3

Session: July 2024 – November 2024 | Program: B.Tech. | Semester: VII

Course Coordinator: Dr. Malvinder Singh Bali

A. Introduction: Department of Data Science & Engineering offers this course to provide students essential knowledge of various components, concepts, applications, mechanisms, and security concerns in the domain of block chain technologies. Due to increased popularity of cryptocurrencies and enhanced security features, block chain technologies are becoming potential candidate to decentralize applications and services in real-world. The course will be beneficial for students to explore and utilize the technologies in development of futuristic data science applications. The course has prerequisites of familiarity with databases, network security and basic programming environments.

B. Course Outcomes: At the end of the course, students will be able to

- [DS4142.1] Describe the key components of a blockchain, including the structure of a block, the Genesis Block, and the concept of linking blocks in the blockchain.
- [DS4142.2] Demonstrate an understanding of cryptographic concepts by explaining how hash functions are used to chain blocks and how digital signatures are utilized to sign transactions in a blockchain.
- [DS4142.3] Apply their knowledge of blockchain technology to perform Simplified Payment Verification (SPV) and construct Merkle Trees for efficient data verification in a distributed ledger.
- [DS4142.4] Analyze and evaluate the concepts of consensus mechanisms, such as Proof-of-Work (PoW) and Proof-of-Stake (PoS), and their implications on the security and efficiency of blockchain networks.
- [DS4142.5] Design and implement a basic smart contract using a programming language like Solidity, showcasing their ability to create decentralized applications on blockchain platforms like Ethereum.

C. Program Outcomes and Program Specific Outcomes:

- [PO.1]. **Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems
- [PO.2]. **Problem Analysis:** Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
- [PO.3]. **Design/Development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
- [PO.4]. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions
- [PO.5]. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations
- [PO.6]. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice
- [PO.7]. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
- [PO.8]. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practices

- [PO.9]. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings
- [PO.10]. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions
- [PO.11]. Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments
- [PO.12]. Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change

At the end of the program, the students will be able to

- [PSO.1].** Understand the role of mathematics, statistics, and AI techniques in the field of data science & engineering.
- [PSO.2].** Apply the acquired knowledge and expertise to perform data analytics tasks for multidimensional data sets.
- [PSO.3].** Develop effective and scalable industrial solutions for real world socio-economic problems using data analytics tools and techniques.

D. Assessment Rubrics:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	MTE (Closed Book)	30
	In class Quizzes and Assignments	30
End Term Exam (Summative)	ETE (Closed Book)	40
	Total	100
Attendance (Formative)	A minimum of 75% Attendance is required to be maintained by a student to be eligible to appear for the ETE. The allowance of 25% includes all types of leaves including medical leaves.	
Homework/ Home Assignment/ Activity Assignment (Formative)	There are situations where a student may have to work in home, especially before a flipped classroom. Some of these works are graded with marks. A student is expected to participate and perform these assignments with full zeal since the activity/ flipped classroom participation by a student will be assessed and marks will be awarded.	

E. Syllabus

Introduction: Structure of a Block, The Genesis Block, Linking Blocks in the Blockchain, Merkle Trees. **Simplified Payment Verification:** Using hash functions to chain blocks, for Proof-of-Work, Digital Signatures to sign transactions, Distributed Ledger, Byzantine Agreement, Eventual Consistency & Bitcoin Consistency- Availability and Partitions, Bitcoin, Smart Contracts, Weak Consistency, Distributed Storage. **Consistent Hashing:** Hypercubic Networks, Mining and Consensus: Decentralized Consensus, Independent Verification of Transactions Mining Nodes, Aggregating Transactions into Blocks, Constructing the Block Header, Successfully Mining the Block, validating a New Block, Assembling and Selecting Chains of Blocks. **Consensus Attacks:** Changing the Consensus Rules, Soft Fork Signaling with Block Version, Consensus Software Development, Ethereum and Bitcoin, block format, mining algorithm, proof-of-stake (PoS) algorithm, account management, contracts and transactions, Solidity language. **Account management:** contracts and transactions, Applications of Blockchain: Case studies

F. References:

1. Andreas M. Antonopoulos, "Mastering Bitcoin: unlocking digital cryptocurrencies", O'Reilly Media, (1e) 2014.
2. Roger Wattenhofer, "Distributed Ledger Technology, The science of the Blockchain", Inverted Forest Publishing, (2e), 2017.
3. Antonopoulos, Andreas M. and Wood, Gavin. "Mastering Ethereum", O'Reilly Media, 2018.
4. George Icahn, "Blockchain: the complete guide to understanding blockchain technology", Amazon publishers, 2017.

G. Lecture Plan:

Lecture No	Topics	Learning Objective	Mode of Delivery	Corresponding CO	Mode of Assessing the Outcome
1	Introduction to Block chain Technology, Evolution	Introduction to the course and target market	Lecture	DS4142.1	Assignments Class Quiz MTE ETE
2	Why Block chain? Advantages/Disadvantages	Basic concepts in blockchain	Lecture	DS4142.1	
3	Components of a Block chain	Key components of blockchain	Lecture	DS4142.1	
4	Introduction to Cryptography, Digital Signature	Knowledge of cryptography concepts and their significance in Block chain technologies	Lecture	DS4142.2	
5	Structure of a Block, The Genesis Block	Basic block components	Lecture	DS4142.1	
6	Linking Blocks in the Blockchain, Demonstration	Blockchain formation through connecting blocks	Lecture	DS4142.1	
7	Hash Function, Merkle Trees	Understanding the transaction Integrity process	Lecture	DS4142.2	
8	Simplified Payment Verification	Decentralized transaction verification process	Lecture	DS4142.2, DS4142.3	Assignments Class Quiz ETE
9	Using hash functions to chain blocks, for Proof-of-Work, Digital Signatures to sign transactions	Mining of blocks from full nodes	Lecture	DS4142.2	
10	Distributed Ledger, Byzantine Agreement	Record keeping and maintenance in blockchain	Lecture	DS4142.3	
11	Introduction to cryptocurrency, Bitcoin	Cryptocurrency Applications of Blockchain	Lecture	DS4142.2	
12	Eventual Consistency & Bitcoin Consistency-Availability and Partitions, CAP Theorem	Consistency of Bitcoin Properties	Lecture	DS4142.3	
13	Smart Contracts, Wallets, Demonstration	Applications of block chain in online contracts	Lecture	DS4142.3	
14	Weak Consistency, Distributed Storage	Storage consistency properties	Lecture	DS4142.3	
15	Introduction to Hashing, Consistent Hashing	Knowledge and properties of Hashing	Lecture	DS4142.2	Assignments Class Quiz MTE ETE
16	Hypercubic Networks	Interactions between full nodes	Lecture	DS4142.2	
17	Mining and Consensus: Decentralized Consensus	Trust building and ledger consistency process in blockchain	Lecture	DS4142.2	
18	Independent Verification of Transactions Mining Nodes, Aggregating Transactions into Blocks	Verification of individual transactions and adding transactions to blocks	Lecture	DS4142.2	
19	Constructing the Block Header, Successfully Mining the Block	Mining individual blocks through Proof of Work	Lecture	DS4142.2, DS4142.4	
20	Validating a New Block	Propagation and validation of blocks from full nodes	Lecture	DS4142.2	
21	Assembling and Selecting Chains of Blocks	Adding validated blocks to the blockchain	Lecture	DS4142.2	
22	Consensus Attacks	Understanding various attacks to block chains i.e 51% attack, double spending attack	Lecture	DS4142.4	Assignments Class Quiz
23	Changing the Consensus Rules	Modifications in rules	Lecture	DS4142.4	

24	Soft Fork Signaling with Block Version	Chain formations in block chains	Lecture	DS4142.4	MTE ETE
25	Consensus Software Development	Understanding the development environment	Lecture	DS4142.4	
26	Ethereum and Bitcoin, comparison	Knowledge of two popular cryptocurrencies	Lecture	DS4142.4	
27	block format, mining algorithm	Block structure and mining algorithms	Lecture	DS4142.4	
28	proof-of-stake (PoS) algorithm	Mining process	Lecture	DS4142.4	
29	account management	User accounts in blockchain	Lecture	DS4142.5	Assignments Class Quiz ETE
30	contracts and transactions	Smart contract execution process	Lecture	DS4142.5	
31	Solidity language	Learning basic programs to create smart contracts	Lecture	DS4142.5	
32	account management	Knowledge of user accounts	Lecture	DS4142.5	
33	contracts and transactions	Understanding process of contract code execution as per triggers generated by transactions	Lecture	DS4142.5	
34	Applications of Blockchain: Case studies	Real life applications of Blockchain other than cryptocurrencies	Lecture	DS4142.5	
35	Applications of Blockchain: Case studies	Real life applications of Blockchain other than cryptocurrencies	Lecture	DS4142.5	
36	Applications of Blockchain: Case studies	Real life applications of Blockchain other than cryptocurrencies	Lecture	DS4142.5	

H. Course Articulation Matrix: (Mapping of COs with POs)

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES												CORRELATION WITH PROGRAM SPECIFIC OUTCOMES		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
DS4142.1	Describe the key components of a blockchain, including the structure of a block, the Genesis Block, and the concept of linking blocks in the blockchain.	3	1	1									2	1		
DS4142.2	Demonstrate an understanding of cryptographic concepts by explaining how hash functions are used to chain blocks and how digital signatures are utilized to sign transactions in a blockchain.	3	2	1	2	1			1	1	1		3	3	1	
DS4142.3	Apply their knowledge of blockchain technology to perform Simplified Payment Verification (SPV) and construct Merkle Trees for efficient data verification in a distributed ledger.	3	3	2	2	2	2		1	1	1		3	2	1	
DS4142.4	Analyze and evaluate the concepts of consensus mechanisms, such as Proof-of-Work (PoW) and Proof-of-Stake (PoS), and their implications on the security and efficiency of blockchain networks.	3	2	1	3	2		1	1	1	1		3	2	1	1
DS4142.5	Design and implement a basic smart contract using a programming language like Solidity, showcasing their ability to create decentralized applications on blockchain platforms like Ethereum.	3	3	2	3	3	2	1	1		1		3	2	1	1

Course Coordinator

Student Representative

Head of the Department